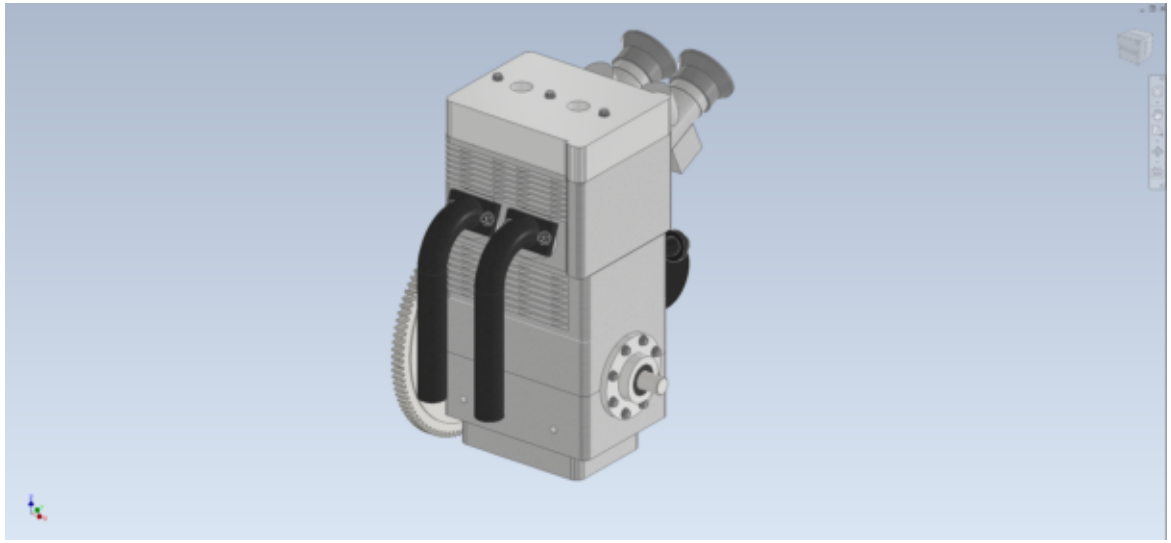
# Two-Cylinder Engine: 296 cc air-cooled DOHC parallel twin, 180 deg crank

TwoCylinderEngine - calculation report. Designed and calculated only: nothing has been built, run or tested. CAD = Inventor, CALC = calculated, FEA = simulated, CAT = catalogue, ASSUM

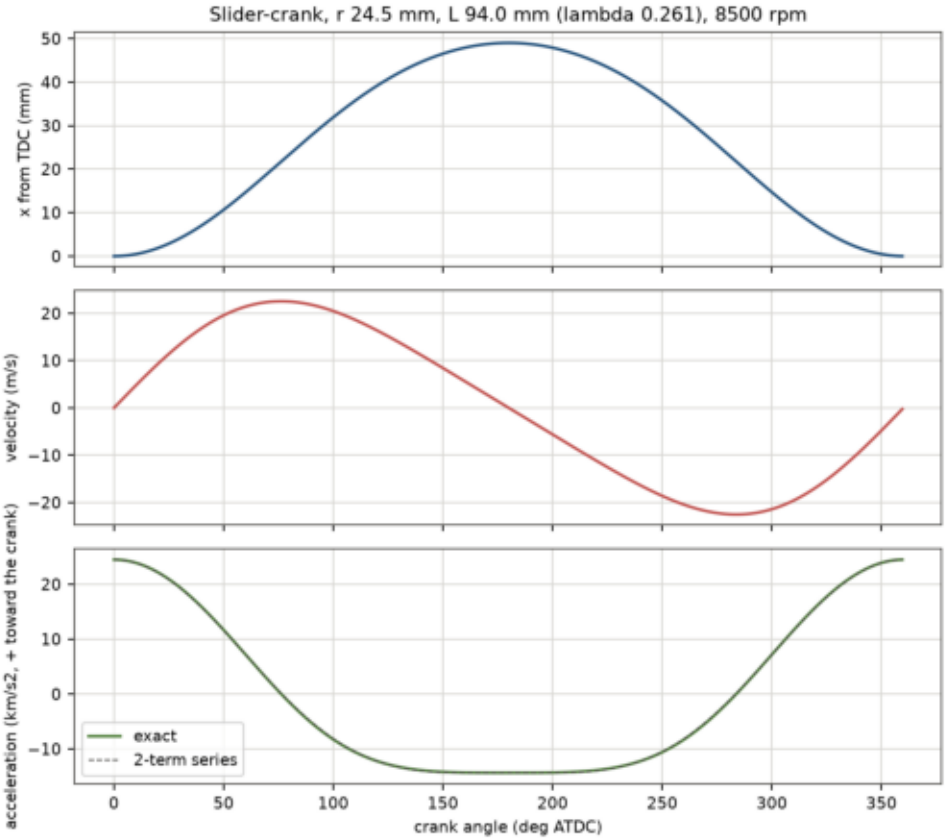


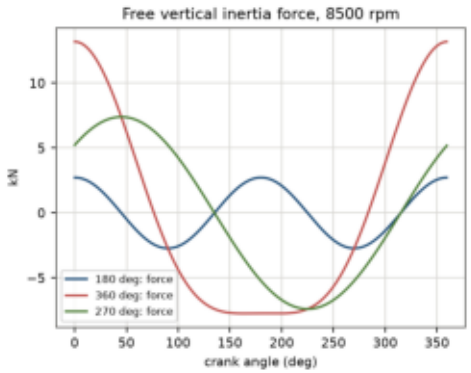
| Item | Value | Source |
|---|---|---|
| Layout | 4-stroke parallel twin, bore x stroke 62 x 49 (Kawasaki EX300 pistons, rings, buckets, springs), rod 94, 295.9 cc, CR 10.6, 180 deg crank (firing 180 / 540) | CALC / CAT |
| Valve train | DOHC, 4 valves / cyl, direct buckets 25; custom 22 / 19 heads; cams 7.15 mm lift, 8th-order polynomial; chain ANSI 25 16T / 32T | CAD |
| Performance | 20.9 kW at 8500 rpm (limit), 24.4 N m at 6500 rpm (single-zone Wiebe model, assumed volumetric efficiency) | CALC |
| Balance | primary force 0 (180 deg), pitch couple 224 N m, secondary force 2726 N at 8500 rpm; balance factor 0.44 | CALC |
| Strength | rod buckling SF 11.8; crankpin press-fit slip SF 2.06; head-stud clamp ratio 5.0; con-rod FEA SF 2.85 (yield) | CALC / FEA |
| Valve train | float 9967 rpm (limit 8500); piston-valve 2.05 / 2.55 mm; in-ex valve 2.64 mm; squish 1.10 mm | CAD |
| Assembly | 195 occurrences, 582 constraints, 0 DOF, 0 clashes at 21 crank angles over 720 deg; driven by one parameter, Crank | CAD |
| Mass | 36.7 kg with starter (32.9 kg without); crank group J 0.0416 kg m2, flywheel J 0.0388 kg m2 | CAD |

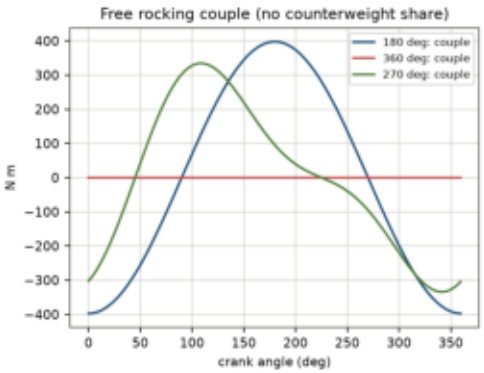
# 1. Kinematics, firing order and balance



Slider-crank with r = 24.5 mm, L = 94 mm (lambda 0.2606). At 8500 rpm: mean piston speed 13.88 m/s, peak 22.54 m/s, acceleration 24,471 m/s2 at TDC and 14,352 m/s2 at BDC (numerical derivative and closed form agree). The Inventor model uses the same expressions as assembly parameters (PinZ1/2, Rod1/2) and cad_props.py compares the measured pin heights, rod angles, cam angles and 8 valve lifts against design.py at 32 crank angles: max error 0.0. Crank layout: a 180 deg crank fires at 180 / 540 deg (uneven) but has no primary shaking force; the 360 deg twin fires evenly but shakes with the full primary force of a single (10,461 N at 8500 rpm), and the 270 deg crank is in between. I chose 180 deg without a balance shaft for a test-stand engine: the free pitch couple (224 N m) and the secondary force (2,726 N) are taken by the stand mounts. Reciprocating mass 0.2694 kg per cylinder (CAD); tungsten slugs in the webs bring the balance factor to 0.44.



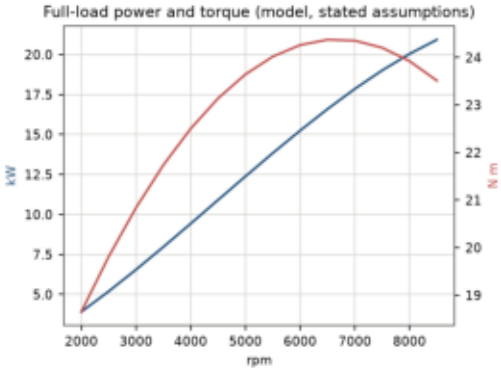



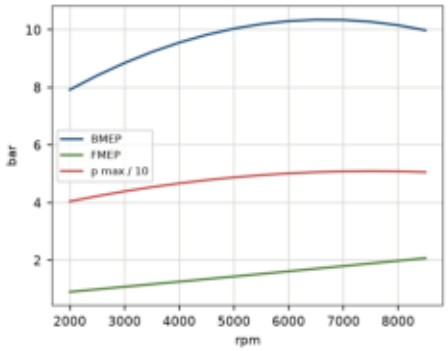



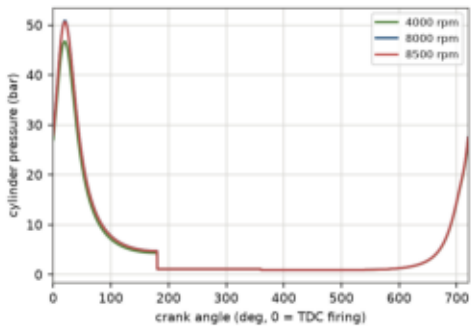
# 2. Cycle model, power and torque

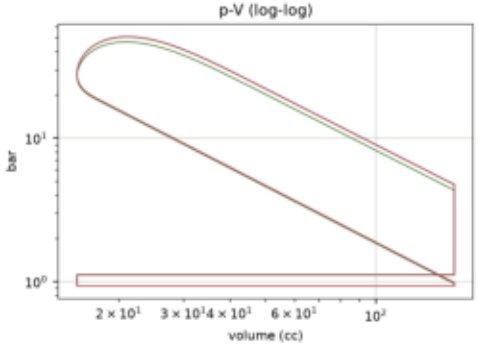


Single-zone cycle: Wiebe heat release (a, m, start, duration = [5.0, 2.0, -14.0, 55.0]), gamma 1.3, AFR 12.8, LHV 44.0 MJ/kg, combustion efficiency 0.95, 80% of the heat to the gas, constant pumping pressures (intake 0.92 bar, exhaust 1.10 bar (constant)). Friction Chen-Flynn 0.4 + 0.005 pmax + 0.09 Sp + 0.0009 Sp^2 bar. Volumetric efficiency is ASSUMED: 0.88 - 0.16 ((n - 7500)/4500)^2 (assumed, no gas dynamics). Inlet Mach index (Taylor) at 8500 rpm Z = 0.48 with the 2 x 22 mm intakes, below the ~0.6 where breathing falls off. Result: 20.9 kW (28 hp) at 8500 rpm and 24.4 N m at 6500 rpm, peak cylinder pressure 51 bar. Reference: Kawasaki EX300 (same bore / stroke, liquid-cooled, fuel-injected, CR 10.6): 39 PS at 11000 rpm, 27 N m at 10000 rpm (Kawasaki service manual); this engine is carburetted and air-cooled, so the lower numbers are expected.



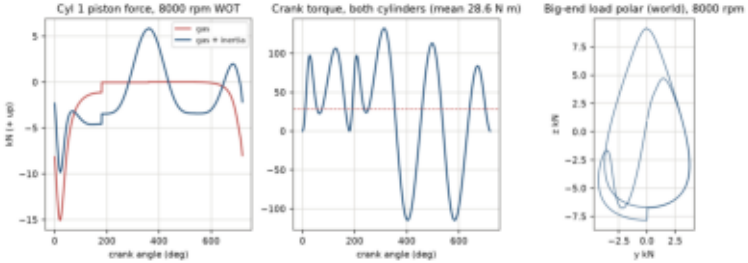



# 3. Loads, bearings and strength

TwoCylinderEngine - calculation report. Designed and calculated only: nothing has been built, run or tested. CAD = Inventor, CALC = calculated, FEA = simulated, CAT = catalogue, ASSUM

| Case | Gas force kN | Rod comp. / tens. kN | Big end max kN | Crank torque N m mean (min..max) |
|---|---|---|---|---|
| WOT 6000 | 14.84 | 11.93 / 3.31 | 10.32 | 28.1 (-65 .. 138) |
| WOT 8000 | 15.05 | 9.85 / 5.86 | 9.14 | 28.6 (-115 .. 132) |
| WOT 8500 (limit) | 14.97 | 9.10 / 6.62 | 10.32 | 28.4 (-130 .. 147) |

| Check | Result | Source |
|---|---|---|
| Con-rod buckling (4340 QT, Johnson) | in plane SF 13.05, out of plane SF 11.84 | CALC |
| Con-rod shank tension | 40.9 MPa at 6.62 kN | CALC |
| Piston pin 16 x 10 | double shear 61.4 MPa, bending 188 MPa (SF 4.79) | CALC |
| Crankpin 28 (52100) | bending 39 MPa (SF 38.89) | CALC |
| Crankpin press fit | first spec 0.040: p 77.5 MPa, slip 186 N m (SF 1.27), web hoop 202 MPa; min fit 0.065: p 126.0 MPa, slip 303 N m (SF 2.06), web hoop 328 MPa; max fit 0.082: p 158.9 MPa, slip 382 N m (SF 2.6), web hoop 413 MPa; peak torque 146.7 N m; mu 0.15 ASSUMED | CALC |
| Head studs M8 10.9 (4 / cyl) | gas 3.97 kN / stud, preload 19.7 kN (25.3 N m), clamp ratio 4.97, hot +4.9 kN -> 674 MPa | CALC |
| Bearing big_end K28x34x17A | WOT 6000: L10 603 h, s0 3.25; WOT 8000: L10 141 h, s0 3.67 (continuous WOT at that speed) | CALC / CAT |
| Bearing main 6306/C3 (2 off) | WOT 6000: L10 1770 h, s0 1.83; WOT 8000: L10 819 h, s0 2.35 (continuous WOT at that speed) | CALC / CAT |

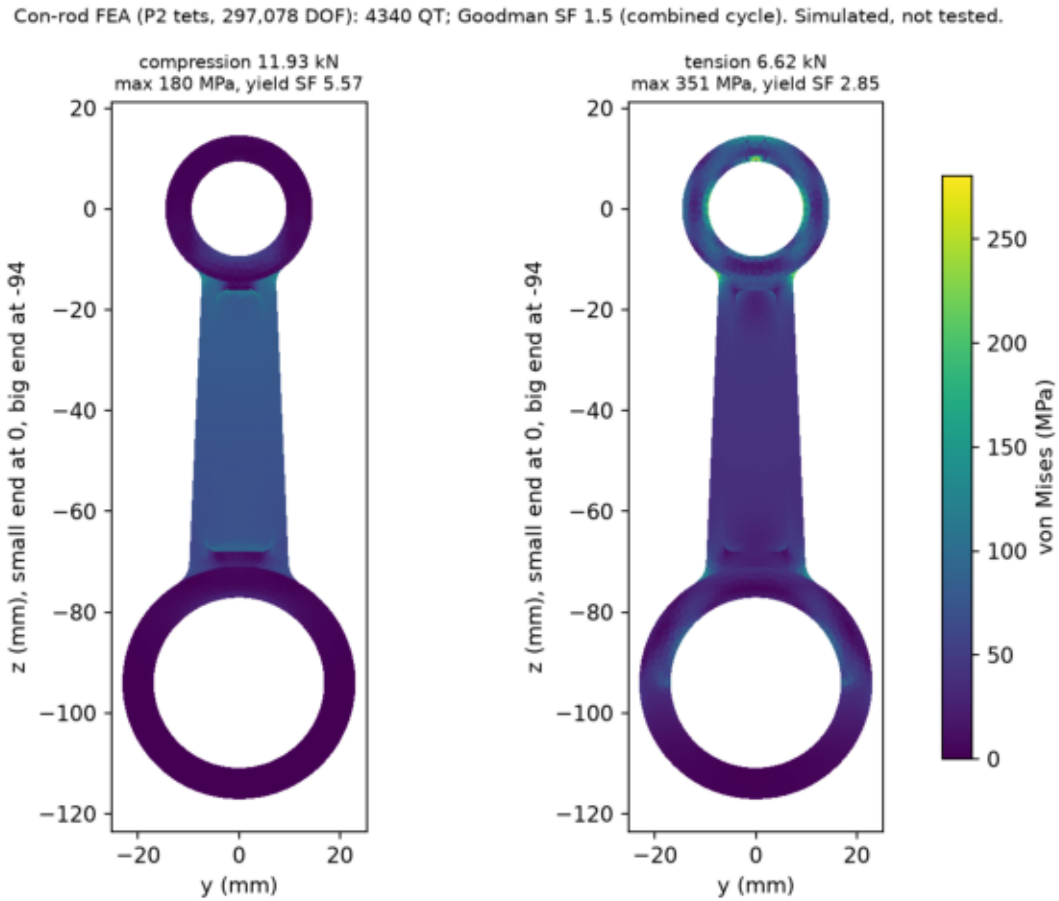



Press-fit respec (2026-11-02): the first 0.04 mm interference gave slip SF 1.27 against the peak crank torque. Web bores are now 27.915 +/-0.004 on the 28 g5 pins (0.065-0.082 mm).

# 4. Con-rod FEA (gmsh + scikit-fem, quadratic tetrahedra)



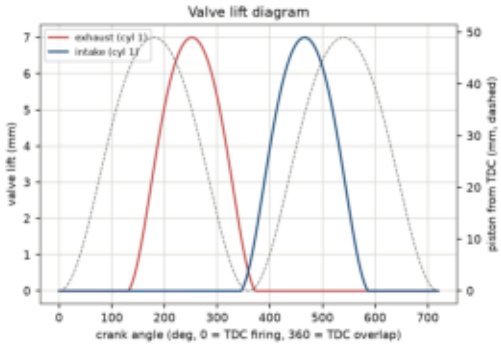
Geometry: the Inventor part exported to STEP, meshed in gmsh at h 1.5 mm (61,305 P2 tetrahedra, 297,078 DOF), solved with scikit-fem and AMG-preconditioned CG. Material 4340 QT: E 205 GPa, Sy 1000, Su 1150 MPa (ASSUMED). Compression 11.93 kN: the pin bears on the lower half of the small-end bore (cosine pressure), the crankpin reacts on the upper half of the big-end bore (fixed; stiffer than a real contact). Tension 6.62 kN: the pin pulls on the top of the small-end eye, reacted on the cap side of the big end. Bronze bush left out (conservative). Results: compression 179.7 MPa max (small-end eye, SF 5.57); tension 350.7 MPa max (small-end eye, SF 2.85). Fatigue (Goodman, signed von Mises, Se 341 MPa): each speed's own tension-compression cycle: WOT 6000: SF 2.37; WOT 8000: SF 1.72; WOT 8500 (limit): SF 1.58. Worst-tension + worst-compression combined (different speeds, very conservative): SF 1.5. Peak stresses sit at the loaded bore skin, where a cosine pressure on a rigid-free bore overstates the local value; a contact model with the bush and a strain-gauged proof test would settle it.

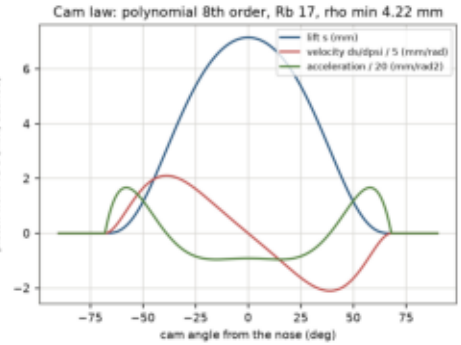


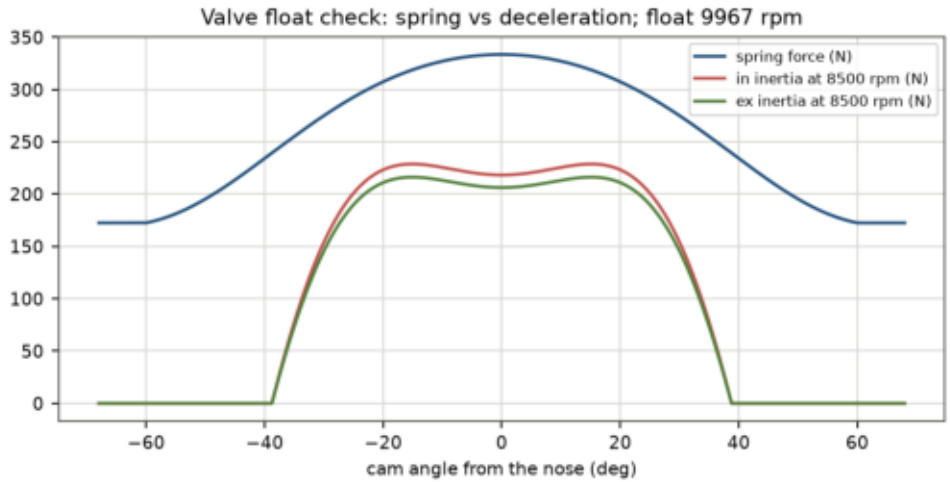
# 5. Valve train: cam law, timing, springs, clearances



Cam: base circle R17, lift 7.15 mm, 8th-order polynomial over +/-68.0 cam deg, flat-faced follower (bucket R12.5): min radius of curvature 4.22 mm (> 0, no undercut), max contact offset 10.52 mm (< bucket radius). Lobe outline checked against the lift law to 6e-05 mm. Valve lift = cam lift - 0.15 mm. Timing (0 mm valve lift): IVO 14 BTDC / IVC 46 ABDC, EVO 48 BBDC / EVC 12 ATDC (240 deg); at 1 mm lift 200 deg; lobe centres 106 ATDC / 108 BTDC. Cams turn with the crank (chain) at half speed. Springs (EX300 single spring, rate 23.0 N/mm ASSUMED): 172 N seat, 334 N open, coil-bind margin 3.25 mm, stress 732 MPa open; moving mass 60.3 / 57.0 g (CAD) -> float 9967 / 10252 rpm. Clearances measured in Inventor through each overlap TDC (2 deg steps): piston to intake valve 2.05 mm at crank 368 (analytic 2.05), to exhaust valve 2.55 mm (analytic 2.55); intake to exhaust valve 2.64 mm; piston to head at TDC 1.10 mm. Respec (2026-11-02): the first layout's 24 / 20.5 mm heads left 0.16 mm between the intake and exhaust valves at overlap; custom 22 / 19 mm heads at an 11.4 mm offset fixed it.



# 6. Flywheel, cooling, working model and assumptions



| Case | Energy swing | Cs with CAD inertia | Target |
|---|---|---|---|
| idle 1500 rpm, 0.30 charge | 100.9 J | 0.0982 | 0.1 |
| WOT 2500 rpm | 190.5 J | 0.0668 | 0.05 |
| WOT 4000 rpm | 210.4 J | 0.0288 | 0.05 |

Cooling (CALC): 20.0 kW to the walls at peak power; fins + blower (25 m/s, h 120.0 W/m2K ASSUMED) and a 4.0 kW oil cooler reject about 9.1 kW, so continuous running is limited to about 9.5 kW; full-power pulls about 78 s for a 40 K rise.

Working model (CAD): 195 occurrences, 582 constraints, 0 DOF, 0 unhealthy constraints; 0 interfering pairs at 21 crank angles over the 720 deg cycle; probes (crank, flywheel, both cams) within 0.0000 deg. A valve spring is excluded only while its valve is open (rigid spring); at full lift each spring overlaps only its own retainer.

Main assumptions: volumetric efficiency curve, Wiebe constants, friction correlation, valve-spring rate (donor spring not measured), piston compression height (donor piston not measured), press-fit friction 0.15, 4340 / 52100 strengths, heat-transfer coefficient. Purchased parts (carburettors, starter, filter, plugs, bearings, needle cages) are envelopes from catalogue dimensions. Prices: $1,705 of the 12,487$ BOM is confirmed on vendor listings; the rest are estimates.